Alzheimer’s Disease Diagnosis Using Machine Learning

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Abstract
Alzheimer’s Disease (AD) is a progressive mental deterioration and incurable neurodegenerative disease, responsible for 60-70% of dementia cases. In addition, Alzheimer’s Disease is known as the fifth-leading cause of death among elderly population. As the global population ages, the prevalence of Alzheimer’s disease is expected to rise significantly, posing a considerable burden on healthcare systems and society as a whole. The objective of this research study is to propose a state-of-the-art and the implementation of various machine learning models to predict Alzheimer’s Disease from a MRI dataset of normal and diseased subject.
Keywords: Alzheimer’s Disease; Machine Learning; Prediction; MRI Dataset

I. Introduction

Alzheimer’s disease (AD) is progressive neurological illness that primarily affects the elderly. It is characterized by memory loss, cognitive decline, and difficulties with normal daily activities. It is the most frequent cause of dementia, responsible for 60-70% of all cases globally. Alzheimer’s disease is predicted to become far more common as the world’s population ages, placing a heavy strain on healthcare institutions and society globally.

The use of machine learning methods to increase the precision and efficiency of alzheimer’s disease diagnosis has gained attention in the last few years. In order to find patterns and biomarkers indicating of Alzheimer’s disease, Machine learning algorithms have the ability to analyze huge amounts of heterogeneous data, including medical evaluations, brain scans (which include positron emission tomography (PET), magnetic resonance imaging (MRI), ,and cerebrospinal fluid (CSF) biomarkers), genetic information, and also useful sensor data.

Nowadays, clinical evaluation which contains cognitive testing, medical records evaluation, and neuroimaging scans is a major component in the diagnosis of Alzheimer disease. But these techniques can be expensive, time consuming, and subjective, which usually results in a delayed diagnosis and start of therapy. Furthermore, with misdiagnosis rates up to 30-40%, clinical diagnostic accuracy continues to be a major difficulty, especially in the early stages of the illness.

The capability of machine learning to combine several data types and identify complex patterns that could be invisible to human eyes is one of the main benefits of utilizing it to diagnose Alzheimer’s disease. Machine learning models can determine the difference between pathological changes related to Alzheimer’s disease and normal aging by training them on huge datasets that include both healthy persons and patients with Alzheimer’s disease.

Lately, a type of deep learning algorithms known as convolution neural networks (CNN) has been gaining popularity. CNNs have shown potential in the analysis of medical imaging data and the diagnosis of a number of diseases,which includes Alzheimer’s disease. Since CNN is able to learn hierarchical features from raw data, they are particularly suitable for tasks involving picture recognition and classification. CNN designs have been effectively used in medical imaging applications including ResNet50, ResNet152, DenseNet, and EfficientNetB7.

ResNet50 is famous for its depth and performance, especially in image classification tasks. The residual connection is an architecture that allows training deeper networks without the problem of vanishing gradients. A number of residual blocks are stacked together in the ResNet50 model. Each residual block consists of convolutional layers followed by shortcut connections. The shortcut connections allow the network to learn residual functions, making it easier to optimize and thus improve accuracy.

ResNet152 is the extension version of the original ResNet50 architecture, introduced by Microsoft Research in 2015. Therefore, it has 152 layers and is designed to deal with even deeper networks effectively, hence addressing the degradation problem encountered in the training of very deep neural networks. The degradation problem means the phenomenon where the accuracy of a deep neural network saturates and degrades rapidly as the network's depth increases. It occurs due to optimization difficulties associated with training deep networks. ResNet152 solves this problem through skip connections or shortcut connections, which allow the network to learn residual functions. These shortcuts allow the network to skip over a few layers so that it could learn the identity function more easily and solve the degradation problem.

A deep learning architecture named DenseNet (Dense Convolutional Network) was proposed in 2017. It provides extensive connections between layers, differentiating it from conventional CNN. Unlike ResNet architecture that uses residual connections, every layer in DenseNet is connected to every other layer in a feed-forward fashion. This architecture pattern provides every layer with direct access to features from any earlier layer, allowing feature reuse and improving gradient flow through the network.

EfficientNetB7 was developed by researchers at Google AI in 2019 as part of the EfficientNet family of CNN. In essence, EfficientNetB7 provides a scalable and efficient method for constructing deep neural networks that attain state-of-the-art performance in many image classification tasks while keeping computational efficiency. More specifically, it was designed to balance the model size, accuracy, and computational cost, thereby being particularly suited for resource-constrained scenarios, like mobile and edge devices. This architecture is highly focused on computational efficiency and scalability and is thus able to achieve competitive performance with fewer parameters compared to ResNet and DenseNet.

Our research will compare the accuracy rates of these CNN architectures for the early prediction of AD. We will compare their performance using benchmark datasets and clinical data to find the most effective way of diagnosing AD. The long-term goal, then, is to provide a reliable and efficient tool for the early detection of AD, which enables the implementation of timely interventions and individualized therapeutic strategies. Our research adds to the advancement of machine learning-based diagnosis of AD and improves patient outcomes in the clinical realm.

II. Related Works

Several studies have proposed AD diagnosis and detection systems that utilize various classification techniques and machine learning methods. Some of the previous studies on Alzheimer's Disease diagnosis are focused on developing models to analyze the anatomical or structural brain images such as MRI and brain functionality to detect any defect or disorder. Furthermore, segmentation tasks were viewed as classification issues, and the voxel, region, or patch-based approaches mostly relied on manually constructed features and feature representations.

In 2021, Xiaomu Tang and Jie Liu proposed Random Forest (RF), Decision Tree (DT), and Support Vector Machine (SVM) algorithms to classify and predict the different disease progress of Alzheimer’s Disease into four groups, such as Cognitive Normal (CN), Early Mild Cognitive Impairment (EMCI), Late Mild Cognitive Impairment (LMCI), and Alzheimer’s Disease (AD). The study was conducted on 560 subjects aged between 60 an 90 who had received an education or participated in work for at least 6 years and met the diagnostic criteria, which was obtained from ADNI (Alzheimer’s Disease Neuroimaging Initiative). The simulation result shows that the proposed RF algorithm had the best prediction effect on different disease processes of AD.

Another study for predicting Alzheimer's Disease was conducted in 2021 by P. C. Muhammed Raees and Vinu Thomas. The study was conducted to predict and classify subjects into three groups, such as Mild Cognitive Impairment (MCI), Alzheimer’s Disease (AD), and Normal classes. A total of 200 subjects diagnosed with early AD, 400 subjects with MCI, and 200 elderly normal control subject were obtained from ADNI dataset of the University of Southern California. The experiments were conducted by utilizing SVM and different models of Deep Neural Network (DNN) algorithms and resulting the highest prediction accuracy of 80-90% for DNN models.

In 2022, Duaa Alsaaed and Samar Fouad Omar proposed convolutional neural network (CNN) model ResNet50 with conventional Softmax, SVM and RF algorithms to detect AD using brain imaging, including MRI and show the parietal atrophy of AD cases. The experiment was conducted on a MRI dataset obtained from ADNI and MIRIAD (Minimal Interval Resonance Imaging in Alzheimer's Disease) containing 549 subjects including subjects with AD and subjects with normal control. The result shows that the model ResNet50 combined with Softmax achieved a higher accuracy than most of the state-of-the-art models.

In 2020, Loris Nanni, Matteo Interlenghi, Sheryl Brahnam, Christian Salvatore, Sergio Papa, Raffaello Nemni, Isabella Castiglioni, and the Alzheimer’s Disease Neuroimaging Initiative (ADNI) proposed CNN, SVM, Transfer-Learning, and Deep Learning approach to compare deep or transfer learning and conventional machine learning when applied to the same neuroimaging studies for the early diagnosis and prognosis of AD. The study was conducted on MRI brain images obtained from ADNI dataset containing more than 600 subjects, including criteria for different diagnostic classes of patients, such as CN subjects, MCI patients, AD patients, Salvatore-509, and Moradi-264. The study shows that a deep learning network trained from scratch on few hundreds of MRI volumes obtained lower performance than either the fusion of conventional machine learning (ML) systems and the ensemble of 2D pre-trained CNNs, due to the limited sample of images used for training.

Another study for detecting AD based on MRI findings using ML models was conducted in 2020 by Gopi Battineni, Nalini Chintalapudi, Francesco Amenta, and Enea Traini. The study was conducted on MRI images from 150 subjects aged 60 and above with fourteen distinct features related to standard AD diagnosis, obtained from Alzheimer's Disease Research Center (ADRC) of Washington University. The study proposed Naive Bayes (NB), Artificial Neural Network (ANN), K-Nearest Neighbor (KNN), and SVM algorithms and the Receiver Operating Characteristic (ROC) curve metric were used to validate the model performance. This study conducted three independent experiments to evaluate each model. In the first experiment, ANN generated the highest accuracy in terms of ROC at 0.812. NB achieved the highest ROC of 0.942 in the second experiment. Last experiment were conducted to combine the four models creating an esemble or hybrid modelling and resulted an improved accuracy of ROC at 0.991.

Gopio Battineni, Mohmmad Amran Hossain, Nalini Chintalapudi, Enea Traini, Venkata rao Dhulipalla, Mariappan Ramasamy, and Francesco Amenta proposed RF, SVM, NB, Logistic Regression (LR), and ensemble learning models, such as Gradient Boosting and AdaBoost to propose a framework based on supervised learning classifiers in the dementia subject categorization as either AD or non-AD based on longitudinal brain MRI features. A total sample of 150 subjects aged between 60 and 96 years of age including demented and non demented subjects were obtained from OASIS (Open Access Series of Imaging Studies) of neurology and patient database was acquired from the longitudinal pool of ADRC at Washington University. The study shows that three classifiers (RF, NB, and Gradient Boosting) achieved the highest average AUC score of 0.98. However, by considering both classification accuracy metric and AUC, the gradient boosting technique can seem a better potential classifier than others.

In 2022, a study to present an AD detection framework consisting of image denoising of an MRI input data set using an adaptive mean filter, preprocessing using histogram equalization, and feature extraction by Haar wavelet transform was conducted by Mustafa Kamar, A. Raghuvira Pratap, Mohd Naved, Abu Sarwar Zamani, P. Nancy, Mahyudin Ritonga, Surendra Kumar Shukla, and F. Sammy. The study proposed LS-SVM-RBF (Least Squares – Support Vector Machine – Radial Basis), SVM, KNN, and RF algorithms for classification conducted on 416 samples obtained from OASIS. The study shows that LS-SVM-RBF algorithm had a higher specificity and accuracy than the other classifiers. The KNN algorithm outperforms the other classifiers in terms of sensitivity and recall.

A study on investigating the usefulness of rule extraction in the assessment of AD using DT and FR algorithms and integrating the extracted rules within an argumentation-based reasoning framework was conducted in 2020 by K. G. Achilleos, S. Leandrou, N. Prentzas, P. A. Kyriacou, A. C. Kakas, and C. S. Pattichis. The study aims to classify subjects into two groups, such as AD subjects and NC subjects. Dataset contains of brain MRI images from a total of 237 subjects aged between 55 and 90 years of age was obtained from ADNI. The study shows that DT classifier had the accuracy of 77%, sensitivity of 66%, and specificity of 88%. RF classifier had the accuracy of 74%, sensitivity of 56%, and specificity of 91%. Argumentation rules had the accuracy of 91%, sensitivity of 87%, and specificity of 95%. Hence, it was proved that augmentation models can be successfully applied.

In 2021, Roobaea Alroobaea, Seifeddine Mechti, Mariem Haoues, Saaed Rubaiee, Anas Ahmed, Murad Andejany, Nicola Luigi Bragazzi, Dilip Kumar Sharma, Bhanu Prakash Kolla, and Sudhakar Sengan conducted common supervised machine learning techniques for automatic AD detection. The experiment was conducted on a total of 2665 CN subjects, 3924 MCI subjects, and 1731 AD subjects which data was obtained from ADNI and OASIS brain dataset. The study shows that the best accuracy values provided by the machine learning classifiers are 99.43% and 99.10% given by respectively, logistic regression and SVM using ADNI dataset, whereas for the OASIS dataset achieved 84.33% and 83.92% given by respectively logistic regression and random forest.

In 2020, a study on Deep Convolutional Neural Network (DCNN) and VGG-16 inspired CNN (VCNN) models implementation to classify the different stages of AD from MRI images was conducted by Ravi Chandaran Suganthe, Rukmani Sevalaiappan Latha, Muthusamy Geetha, and Gobichettipalayam Ramakrishnan Sreekanth. The experiments were carried out on an ADNI dataset containing 1000 sample brain images, which include the 250 images from CN subjects, 250 images from EMCI subjects, 250 images from LMCI subjects, and 250 images from AD subjects. The study shows that more than 90% of accuracy was achieved in both the models, and VCNN produces higher accuracy.

III. Material and Methods

Dataset

The dataset used in this research study have been obtained from Kaggle, an online community of data scientists and machine learning practitioners under Google LLC. Consisting a total of 6400 preprocessed magnetic resonance imaging (MRI) images, the dataset has four classed of images: Mild Demented (896 images), Moderate Demented (64 images), Non Demented (3200 images), and Very Mild Demented (2240 images). The MRI images visualize the axial planes of brain anatomy. These images were collected from several websites, hospitals, and public repositories, including ADNI (Alzheimer's Disease Neuroimaging Initiative), a non-profit organization launched in 2003 by the National Institute of Biomedical Imaging and Bioengineering.

Method

The main purpose of this research is to compare the accuracy rates of various CNN-based architectures to predict Alzheimer's Disease from a MRI dataset of normal and diseased subject. Pre-trained CNN deep learning models ResNet50, ResNet152, DenseNet, and EfficientNetB7 were utilized as automated technique for extracting features from MRI images to diagnose Alzheimer's Disease. The accuracy of these CNN models were evaluated using metric evaluation.

3.1. Data Collection

The dataset used in this research study have been obtained from Kaggle, an online community of data scientists and machine learning practitioners under Google LLC. Collected from several websites, hospitals, and public repositories, the dataset consists of preprocessed magnetic resonance imaging (MRI) images. Consisting a total of 6400 MRI images, the dataset has four classed of images: Mild Demented (896 images), Moderate Demented (64 images), Non Demented (3200 images), and Very Mild Demented (2240 images). 80% of the dataset are used as training dataset, and the remaining 20% are used as testing dataset.

3.2. Implementation of Machine Learning Method

After collecting data, machine learning methods are implemented. Data pre-processing phase of the MRI is not executed because the dataset contains pre-processed MRI images. The data pre-processing phase objective is to transform the data into a more optimal representation to match the pre-trained CNN's input size requirements. In this case, the skull from the MRI images was removed to extract the brain and noise was eliminated to improve the model performance. Then, smoothing techniques were used to reduce noise within an image and produce a less pixelated image.

The method implemented for image recognition and processing is Convolutional Neural Network (CNN). CNN methodology contains several types of layers that are commonly used to extract features and make predictions, such as Input Layer, Convolutional Layer, Activation Layer, Pooling Layer, Batch Normalization Layer, Dropout Layer, and Fully Connected Layer. Input layer is used to receive the input data, which in this case the inputted data is the pixel values of image inputted.

Convolutional layer is the essential part and the core building block of deep learning CNN. Its task is to perform feature extraction process, resulting output sets of 2D matrices called feature maps. Each convolutional layer consists of a fixed number of filters that act as feature detectors and extract the features by convolving inputted image with these filters. Each filter acquires the ability to detect the analyzed image's low level features such as colors, edges, blobs, and corners during training process.

Activation layer introduce non-linearity to the network, which allows it to learn complex patterns and relationships in the data. Pooling layer has sub-sampling layer, responsible for decreasing the size of the feature maps that produced by the convolutional layers. Max pooling is the most used pooling operation to reduce the feature maps by reducing the small region in the image with the maximum value in the region. Max pooling process is performed to avoid overfitting by providing an abstract of the image representation regions and minimizes the computational cost by decreasing the number of parameters. Overfitting is an undesirable machine learning behavior that occurs when the machine learning model gives accurate predictions for training data but now for new data.

Batch normalization layer is used to normalize the convolution layer's output, features maps, by setting the batch's average to 0 and the variance to 1. This layer speeds up the training process, using higher learning rates. It prevents the gradients of the model from vanishing during backpropagation. Deep learning models with batch normalization layers are more robust against improper weights initialization.

Dropout layer is used to avoid overfitting phenomena. During the training, the mechanism of this technique is removing neurons randomly. The number of removed neurons are controlled by dropout rate parameter (decides the likelihood of neuron removal). The neurons are removed only during the training process.

Fully connected layer is the last layer of convolutional neural network and acts as a classifier. Its function is to connect the layers in the network and give the final result of the classification. Usually, it is followed by the final layer with a normalized exponential function (Softmax).

In this study, CNN architecture used for the pre-trained CNN model are ResNet50, ResNet152, DenseNet, and EfficientNetB7. These CNN architecture models share common layers such as convolutional, pooling, normalization, and fully connected layers. Regardless having common layers, these CNN architecture models have distinct differences such as the type of blocks used and the use of bottleneck and transition layers.

ResNet50 and ResNet152 CNN models utilize residual blocks to mitigate the vanishing gradient problem, allowing the training of very deep networks. DenseNet model utilizes dense blocks, where each layer receives input from all preceding layer. This dense connectivity patterns allows feature reuse and propagation. EfficientNetB7 model utilizes efficient building blocks inspired by MobileNetV2 architecture, a lightweight CNN model introduced in 2019 by Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen.

ResNet152 and EfficientNetB7 models incorporate bottleneck layers within deeper layers in order to improve computational efficiency. DenseNet model also incorporates bottleneck layers within dense blocks to reduce computational complexity. However, ResNet50 model does not incorporate bottleneck layers.

DenseNet incorporates transition layers to control the growth of feature map dimensions and improve model efficiency. However, transition layers are not incorporated in ResNet50 and ResNet152 architectures. EfficientNetB7 incorporates compound scaling to simultaneously scale up the network depth, width, and resolution, achieving improved performance without significantly increasing computational cost.

3.3 MRI Image Classification

Classification process can be done by fully connected layer from ResNet50 network, or can be replaced by another classifiers, such as Support Vector Machine and Random Forest classifier. In this study, CNN models will be evaluated with Softmax classifier.

Softmax classification layer is a part of CNN architecture ResNet50 used to classify labelled data and calculate the probability of each ground-truth label of outputs between 0 and 1, and output values converted to perceptible values.

$$f(x)i = \frac{e^{zj}}{\sum_{n=1}^{N} e^{zk}} for\ j = 1, \ldots\ldots, N,$$

N is denoted as the dimension of random values (x), which are converted to the meaningful values between 0 and 1 by Softmax function f(x).

3.4 Performance Evaluation

Performance indicators are used to evaluate ResNet50, ResNet152, DenseNet and EfficientNetB7 models performance. Performance indicators used in this study are accuracy, precision, recall, and F1-score. True positives (TP) refer to the classifier's positive tuples that were correctly labelled. False positives (FP) are the negative tuples that were incorrectly labelled as positive. True negative (TN) are the negative tuples that the classifier correctly labelled. False negatives (FN) are the positive tuples that were mislabelled as negative.

Accuracy (ACC) is the percentage of the number of records classified correctly versus the total records. The equation is shown as below.

$$ACC = (TP + TN) / (TP + TN + FP + FN)$$

Precision (PRE) measures the accuracy of positive predictions. The equation is shown as below.

$$PRE = TP / (TP + FP)$$

Recall (REC) measure the ability of the classifier to find all the positive samples. The equation is shown as below.

$$REC = TP / (TP + FN)$$

F1-Score (F1) is the harmonic mean of precision and recall. It provides a balance between precision and recall. The equation is shown as below.

$$F1 = 2 * PRE * REC / (PRE + REC)$$

Recall (REC) measure the ability of the classifier to find all the positive samples. The equation is shown as below.

$$REC = TP / (TP + FN)$$

IV. Results & Discussion

For this research, MRI image from each AD category were randomly selected as the test subject. The selected MRI images from each AD category, which are Non Demented, Very Mild Demented, Mild Demented, and Moderate Demented were used as inputted data for the ResNet50, ResNet152, DenseNet, and EfficientNetB7 CNN model architecture code.

The simulations conducted using ResNet50 architecture achieved an average accuracy of 35.25%, average precision of 58.75%, average recall of 35.25%, and average F1-score of 38.50%

ResNet50					
No	Class	Accuracy	Precision	Recall	F1-Score
1	Non-Demented	36%	59%	36%	39%
2	Very Mild Demented	35%	58%	35%	38%
3	Mild Demented	36%	59%	36%	39%
4	Moderate Demented	34%	59%	34%	38%
Average		35,25%	58,75%	35,25%	38,50%

Table 1.1 Resnet50 Model Evaluation Results

The simulations conducted using ResNet152 architecture achieved an average accuracy of 61.50%, average precision of 65.25%, average recall of 61.50%, and average F1-score of 60.00%

ResNet152					
No	Class	Accuracy	Precision	Recall	F1-Score
1	Non-Demented	62%	66%	62%	60%
2	Very Mild Demented	61%	65%	61%	59%
3	Mild Demented	61%	65%	61%	60%
4	Moderate Demented	62%	65%	62%	61%
Average		61,50%	65,25%	61,50%	60,00%

Table 1.2 Resnet152 Model Evaluation Results

The simulations conducted using DenseNet architecture achieved an average accuracy of 63.75%, average precision of 76.50%, average recall of 63.75%, and average F1-score of 62.75%

DenseNet					
No	Class	Accuracy	Precision	Recall	F1-Score
1	Non-Demented	65%	76%	65%	64%
2	Very Mild Demented	63%	77%	63%	62%
3	Mild Demented	64%	77%	64%	63%
4	Moderate Demented	63%	76%	63%	62%
Average		63,75%	76,50%	63,75%	62,75%

Table 1.3 DenseNet Model Evaluation Results

The simulations conducted using EfficientNetB7 architecture achieved an average accuracy of 64.25%, average precision of 78.00 %, average recall of 64.25%, and average F1-score of 64.00%

EfficientNetB7					
No	Class	Accuracy	Precision	Recall	F1-Score
1	Non-Demented	65%	79%	65%	65%
2	Very Mild Demented	64%	78%	64%	63%
3	Mild Demented	64%	77%	64%	64%
4	Moderate Demented	64%	78%	64%	64%
Average		64,25%	78,00%	64,25%	64,00%

Table 1.4 EfficientNetB7

Comparing the four pre-trained convolutional neural network (CNN) architectures used in the simulation, the highest accuracy was achieved by EfficientNetB7 architecture model with the average accuracy of 64.25%, average precision of 78.00 %, average recall of 64.25%, and average F1-score of 64.00%. The lowest accuracy was achieved by ResNet50 with the average accuracy of 35.25%, average precision of 58.75%, average recall of 35.25%, and average F1-score of 38.50%. Accuracy defines the percentage of the number of records classified correctly. Precision defines the accuracy of positive predictions. Recall defines the ability of the classifier to find all positive samples and F1 Score is the harmonic mean of precision and recall which provides a balance between precision and recall. Hence, it can be concluded that the EfficientNetB7 CNN architecture model is the most effective model to diagnose Alzheimer's Disease.

Table of Comparison					
No	CNN Architecture	Accuracy	Precision	Recall	F1-Score
1	ResNet50	35,25%	58,75%	35,25%	38,50%
2	ResNet152	61,50%	65,25%	61,50%	60,00%
3	DenseNet	63,75%	76,50%	63,75%	62,75%
4	**EfficientNetB7**	**64,25%**	**78,00%**	**64,25%**	**64,00%**

Table 1.5 Table of Model Comparisons

V. Conclusion

It can be concluded that EfficientNetB7 emerged as the strongest performer across all metrics (accuracy, precision, recall, and F1-score). This is likely due to its ability to achieve high accuracy with a relatively low number of parameters, potentially making it more efficient to train and deploy compared to deeper models like ResNet152. ResNet152 achieved better accuracy than ResNet50, demonstrating the benefit of increased depth for this image classification task. However, it did not outperform EfficientNetB7 or DenseNet in terms of precision and F1-score. DenseNet offered a competitive performance with accuracy and F1-score close to EfficientNetB7. However, EfficientNetB7 surpassed DenseNet in precision. ResNet50 lagged behind the other models in most metrics. This could be due to its shallower architecture compared to the other models.

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